CONVERGENT CHAIN-OF-THOUGHT BEHAVIOR IN LLM PROOF VERIFICATION AND ITS CORRELATION WITH PROOF COMPLEXITY

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Abstract. We investigate the relationship between classical proof complexity measures and the convergence behavior of chain-of-thought (CoT) reasoning in large language model (LLM) proof verification. We model CoT verification as a stochastic saturation process whose parameters depend on proof structure, and we analyze how convergence rate, asymptotic accuracy, and required test-time compute relate to the length, width, depth, and space of resolution proofs. Our main results are: (1) a rigorous error accumulation bound showing that asymptotic verification accuracy decays exponentially in proof length; (2) a proof that self-consistency amplification via majority voting requires $\Omega(1/(p-\frac{1}{2})^2)$ independent chains to reach a target accuracy, where the base accuracy p itself depends on proof complexity; and (3) a convergence rate analysis showing that the number of CoT steps needed to reach ε-convergence scales as $O(\log(1/\varepsilon)/\lambda)$, where λ is inversely related to a composite complexity measure. We validate these theoretical predictions computationally on 129 synthetic proof instances spanning four well-studied formula families (Tseitin, pigeonhole, pebbling, random k-CNF), obtaining Spearman rank correlations exceeding -0.99 between the composite complexity measure and convergence rate. The framework yields testable predictions connecting proof complexity theory to the scaling of test-time compute in neural proof verification.

1. Introduction

Large language models have demonstrated remarkable mathematical reasoning capabilities when augmented with chain-of-thought (CoT) prompting, in which a model produces a sequence of intermediate reasoning steps before arriving at a final answer [12]. A natural application is *proof verification*: given a purported proof of a mathematical statement, a model checks correctness step by step. Recent work shows that verification accuracy can be substantially improved by generating multiple CoT traces and aggregating their judgments via majority voting—a strategy known as *self-consistency* [11]—and that the returns from additional test-time compute follow difficulty-dependent saturation curves [9, 8].

A central question, however, remains open: *what structural properties of a proof determine the difficulty of verifying it?* Snell et al. [9] characterize difficulty using the model-dependent pass@1 rate, while Shi and Jin [8] demonstrate that verification difficulty does not simply track problem-solving difficulty. Neither framework identifies proof-intrinsic features that predict verification convergence a priori.

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Proof complexity theory offers a rich collection of such features. The *length* (number of inference steps), *width* (maximum clause size), *depth* (height of the proof DAG), and *space* (maximum simultaneously stored clauses) of resolution proofs are well studied and exhibit deep interrelationships [1, 2]. Jarvisaälo et al. [4] showed that space complexity correlates with practical SAT solver running time more strongly than length or width. Yet no prior work connects these measures to LLM verification behavior.

In this paper we bridge proof complexity theory and CoT verification convergence. We model CoT verification as a stochastic saturation process whose parameters are governed by proof structure, and we prove rigorous bounds relating proof complexity to verification accuracy and required compute. Our contributions are as follows.

- We formalize CoT verification convergence as an exponential saturation process and prove that the convergence step scales as $O(\log(1/\varepsilon)/\lambda)$, where λ depends inversely on a composite proof complexity measure (Theorem 3.3).
- We establish an error accumulation bound showing that asymptotic verification accuracy decays exponentially in proof length under a step-independence assumption (Theorem 3.6).
- We prove that self-consistency amplification via majority voting requires $\Theta(1/(p-\frac{1}{2})^2)$ independent chains, connecting proof complexity to test-time compute scaling (Theorem 3.10).
- We validate these predictions computationally on 129 synthetic proof instances across four formula families, obtaining strong rank correlations ($\rho = -0.99$) between the composite complexity measure and convergence rate. We identify a phase transition in verification difficulty at a critical complexity threshold (Proposition 3.15).

The paper is organized as follows. Section 2 introduces the necessary background from proof complexity theory and CoT reasoning. Section 3 presents our main results with complete proofs and computational verification. Section 4 discusses connections to prior work, implications, and open questions. Section 5 concludes.

2. Preliminaries

We recall the relevant notions from propositional proof complexity and chain-of-thought reasoning, and fix notation used throughout the paper.

2.1. Propositional proof complexity.

A *literal* is a Boolean variable x or its negation $\bar{x}$. A *clause* is a disjunction of literals. A *conjunctive normal form* (CNF) formula F is a conjunction of clauses. We write $\mathrm{Var}(F)$ for the set of variables appearing in F and set $n = |\mathrm{Var}(F)|$.

Definition 2.1 (Resolution proof system)**.** The *resolution* rule derives a new clause from two parent clauses sharing a complementary literal:

$$\frac{C \vee x \qquad D \vee \bar{x}}{C \vee D}.$$

A *resolution refutation* (or *proof*) of an unsatisfiable CNF formula F is a sequence of clauses $C_1, C_2, \ldots, C_S$ such that each C_i is either a clause of F or is derived from two earlier clauses by the resolution rule, and C_S is the empty clause $\perp$.

Definition 2.2 (Proof complexity measures)**.** Let $\pi = (C_1, \ldots, C_S)$ be a resolution refutation of F.

(i) The *length* (or *size*) of π is $S(\pi) = S$.
(ii) The *width* of π is $w(\pi) = \max_{1 \le i \le S} |C_i|$, where $|C_i|$ denotes the number of literals in C_i.
(iii) The *depth* of π is the height of its derivation DAG.
(iv) The *space* of π is the maximum number of clauses that must be simultaneously retained in memory when executing the proof in a sequential blackboard model.

We write $S(F)$, $w(F \vdash \bot)$, $d(F)$, and $\mathrm{Sp}(F)$ for the respective minima over all refutations of F.

The following classical results relate these measures.

Theorem 2.3 (Ben-Sasson and Wigderson [1])**.** *For any unsatisfiable CNF formula F on n variables with initial clause width $w(F)$,*

$$w(F \vdash \bot) \le w(F) + O\Big(\sqrt{n \ln S(F)}\Big).$$

Equivalently, $S(F) \ge \exp\Big(\Omega\Big(\frac{(w(F \vdash \bot) - w(F))^2}{n}\Big)\Big)$.

We use the following well-known formula families as test instances.

Definition 2.4 (Tseitin formulas)**.** Let $G = (V, E)$ be a connected graph and $\sigma\colon V \to \{0,1\}$ a labeling with $\sum_{v \in V} \sigma(v) \equiv 1 \pmod 2$. The *Tseitin formula* $\mathrm{Ts}(G, \sigma)$ introduces a variable x_e for each edge $e \in E$ and, for each vertex v, a set of clauses encoding $\bigoplus_{e \ni v} x_e = \sigma(v)$. These formulas are always unsatisfiable because the total parity constraint is odd.

Definition 2.5 (Pigeonhole formulas)**.** The *pigeonhole principle* formula PHP_n^m (with $m > n$) has variables $p_{i,j}$ for $1 \le i \le m$, $1 \le j \le n$, with clauses encoding "each pigeon maps to at least one hole" and "no two pigeons share a hole." Haken [2] proved that $S(\mathrm{PHP}_n^{n+1}) = 2^{\Omega(n)}$.

Definition 2.6 (Pebbling formulas)**.** Let $G = (V, E)$ be a directed acyclic graph with a unique sink t. The *pebbling formula* $\mathrm{Peb}(G)$ has a variable x_v for each $v \in V$, with clauses encoding: (i) source vertices are true; (ii) if all predecessors of v are true then v is true; and (iii) the sink is false. The space complexity of $\mathrm{Peb}(G)$ equals the black pebbling number of G, plus a constant [1].

2.2. **Chain-of-thought verification.** We model chain-of-thought verification as a sequential stochastic process. An LLM verifier receives a proof and processes it over discrete *reasoning steps* $t = 1, 2, \ldots$ At each step, the verifier updates its assessment of the proof's correctness.

Definition 2.7 (Process reward model)**.** A *process reward model* (PRM) assigns a correctness probability $r_t \in [0,1]$ to each step t of a solution or proof trace. The solution-level score is $\prod_{t=1}^{T} r_t$, where T is the total number of steps [5].

Definition 2.8 (Self-consistency)**.** Given N independent CoT verification traces for a proof F, each producing a binary judgment $Y_i \in \{0,1\}$ with $\Pr[Y_i = 1] = p$,

the *self-consistency* (majority vote) decision is

$$\hat{Y}_N = \mathbf{1}\left[\sum_{i=1}^{N} Y_i > N/2\right].$$

When $p > 1/2$, the majority vote accuracy exceeds p and converges to 1 as $N \to \infty$ [11].

2.3. **Notation.** Throughout the paper, F denotes an unsatisfiable CNF formula, $S(F)$ its minimum resolution proof length, $w(F \vdash \bot)$ its minimum proof width, $d(F)$ its minimum proof depth, and $\mathrm{Sp}(F)$ its minimum proof space. We write ln for the natural logarithm and log for the base-2 logarithm. The notation $f = O(g)$, $f = \Omega(g)$, and $f = \Theta(g)$ carries the standard meaning. The Spearman rank correlation coefficient is denoted ρ. All probabilities are with respect to the randomness in the CoT generation process.

3. Main results

We present our framework in three parts. First, we formalize the saturation model for CoT verification and analyze its convergence properties. Second, we prove bounds connecting proof complexity to verification accuracy and required compute. Third, we describe computational experiments validating the theoretical predictions.

3.1. **The saturation model.** We model the accuracy of a CoT verifier as an exponential saturation process.

Definition 3.1 (CoT verification process)**.** For an unsatisfiable CNF formula F, the *CoT verification accuracy* at reasoning step t is

$$p_F(t) = p_{\max}(F) - \big(p_{\max}(F) - p_0(F)\big) \cdot e^{-\lambda(F)\cdot t}, \tag{3.1}$$

where:

(i) $p_0(F) \in (0.5, 1)$ is the *initial accuracy* (accuracy from a single-step assessment);
(ii) $p_{\max}(F) \in (p_0(F), 1)$ is the *asymptotic accuracy ceiling*;
(iii) $\lambda(F) > 0$ is the *convergence rate*.

The saturation form (3.1) is motivated by the empirical observation that verification accuracy follows diminishing-returns curves as a function of test-time compute [9, 8].

Definition 3.2 (Convergence step)**.** For $\varepsilon > 0$, the *ε-convergence step* of F is

$$\tau_F(\varepsilon) = \min\big\{t \in \mathbb{N} : p_{\max}(F) - p_F(t) < \varepsilon\big\}.$$

Theorem 3.3 (Convergence rate)**.** *Let F be an unsatisfiable CNF formula and let $p_F(t)$ be defined by* (3.1) *with parameters $p_0(F)$, $p_{\max}(F)$, and $\lambda(F) > 0$. Then:*

(a) $p_F(t)$ is strictly increasing and $\lim_{t\to\infty} p_F(t) = p_{\max}(F)$.
(b) The ε-convergence step satisfies

$$\tau_F(\varepsilon) = \left\lceil \frac{1}{\lambda(F)} \ln\left(\frac{p_{\max}(F) - p_0(F)}{\varepsilon}\right)\right\rceil. \tag{3.2}$$

(c) If $\lambda(F) = \lambda_0/(1 + C(F))$ for some composite complexity measure $C(F) \geq 0$ and constant $\lambda_0 > 0$, then $\tau_F(\varepsilon)$ is increasing in $C(F)$.

Proof. **Part (a).** We have $p'_F(t) = (p_{\max}(F) - p_0(F)) \cdot \lambda(F) \cdot e^{-\lambda(F)t}$. Since $p_{\max}(F) > p_0(F)$ and $\lambda(F) > 0$, the derivative is strictly positive for all $t \geq 0$. The limit follows because $e^{-\lambda(F)t} \to 0$.

Part (b). The gap at step t is

$$p_{\max}(F) - p_F(t) = (p_{\max}(F) - p_0(F)) \cdot e^{-\lambda(F)t}.$$

Setting this less than ε and solving for t gives $t > \frac{1}{\lambda(F)} \ln\left(\frac{p_{\max}(F)-p_0(F)}{\varepsilon}\right)$. Taking the ceiling yields (3.2).

Part (c). Under the stated form, $\tau_F(\varepsilon) = \lceil \frac{1+C(F)}{\lambda_0} \ln\left(\frac{p_{\max}(F)-p_0(F)}{\varepsilon}\right) \rceil$. Since $1 + C(F)$ is increasing in $C(F)$ and the logarithmic factor is positive, $\tau_F(\varepsilon)$ is increasing in $C(F)$. □

We now specify how the convergence rate depends on proof structure.

Definition 3.4 (Composite complexity measure)**.** For an unsatisfiable CNF formula F, the *composite complexity measure* is

$$C(F) = \log S(F) + \log w(F \vdash \bot) + \log d(F), \tag{3.3}$$

where $S(F)$, $w(F \vdash \bot)$, and $d(F)$ are the minimum proof length, width, and depth, respectively.

Remark 3.5. The choice of a sum of logarithms corresponds to taking the logarithm of the product $S(F) \cdot w(F \vdash \bot) \cdot d(F)$. This multiplicative combination captures the intuition that proof verification difficulty grows with each dimension of proof complexity. The logarithmic scaling ensures that the composite measure is comparable across formula families with vastly different parameter ranges.

3.2. **Error accumulation and accuracy bounds.** We now prove that proof length governs the asymptotic accuracy ceiling through an error accumulation mechanism.

Theorem 3.6 (Error accumulation bound)**.** *Let F be an unsatisfiable CNF formula with minimum proof length $S(F)$. Suppose that a CoT verifier checks each proof step independently, accepting step i correctly with probability at least $1-\varepsilon_0$ for some $\varepsilon_0 \in (0, 1/2)$. Then the probability that the verifier correctly identifies the proof as valid satisfies*

$$p_{\max}(F) \leq (1 - \varepsilon_0)^{S(F)} \leq e^{-\varepsilon_0 \cdot S(F)}. \tag{3.4}$$

Proof. Under the step-independence assumption, the verifier accepts the proof as valid if and only if it accepts every step. Since each step is accepted correctly with probability at most $1 - \varepsilon_0$, the probability that all $S(F)$ steps are accepted correctly is at most $(1 - \varepsilon_0)^{S(F)}$. The second inequality follows from $1 - x \leq e^{-x}$ for all $x \geq 0$. □

Corollary 3.7. *Under the hypotheses of Theorem 3.6, if $S(F) \geq \ln(1/\delta)/\varepsilon_0$, then $p_{\max}(F) \leq \delta$.*

Proof. From (3.4), $p_{\max}(F) \leq e^{-\varepsilon_0 \cdot S(F)} \leq e^{-\ln(1/\delta)} = \delta$. □

Remark 3.8. The step-independence assumption is a simplification. In practice, errors at one step may be correlated with errors at subsequent steps—for example, because the CoT trace conditions on prior reasoning. Ton et al. [10] show that once a model encounters an unidentifiable sub-task, all subsequent steps contribute zero

information, which is a stronger form of error propagation. Theorem 3.6 therefore provides an *optimistic* bound; the true accuracy degradation with proof length may be worse.

Example 3.9. Consider the pigeonhole formula PHP_n^{n+1}. By Haken [2], $S(\mathrm{PHP}_n^{n+1}) = 2^{\Omega(n)}$. Even with a highly accurate per-step verifier having $\varepsilon_0 = 0.001$, Theorem 3.6 gives $p_{\max} \le e^{-0.001 \cdot 2^{\Omega(n)}}$, which drops below any constant for sufficiently large n. This shows that pigeonhole formulas present a fundamental barrier for step-by-step verification.

3.3. **Self-consistency amplification.** We analyze how majority voting over independent chains can boost verification accuracy, and connect the required number of chains to proof complexity.

Theorem 3.10 (Self-consistency amplification)**.** *Let $Y_1, \ldots, Y_N$ be independent Bernoulli random variables with common parameter $p > 1/2$. The majority vote $\hat{Y}_N = \mathbf{1}[\sum_{i=1}^N Y_i > N/2]$ satisfies:*

(a) $\Pr[\hat{Y}_N = 1] \ge 1 - \exp\bigl(-2N(p - \frac{1}{2})^2\bigr)$.

(b) To achieve $\Pr[\hat{Y}_N = 1] \ge 1 - \delta$*, it suffices to take*

$$N \ge \frac{\ln(1/\delta)}{2(p - \frac{1}{2})^2}. \tag{3.5}$$

Proof. **Part (a).** Let $\bar{Y} = \frac{1}{N}\sum_{i=1}^N Y_i$. The majority vote errs when $\bar{Y} \le 1/2$. By Hoeffding's inequality [3], for any $t > 0$,

$$\Pr\bigl[\bar{Y} - p \le -t\bigr] \le \exp(-2Nt^2).$$

Setting $t = p - 1/2 > 0$, we obtain

$$\Pr\bigl[\bar{Y} \le \tfrac{1}{2}\bigr] \le \Pr\bigl[\bar{Y} - p \le -(p - \tfrac{1}{2})\bigr] \le \exp\bigl(-2N(p - \tfrac{1}{2})^2\bigr).$$

Therefore $\Pr[\hat{Y}_N = 1] \ge 1 - \exp\bigl(-2N(p - \frac{1}{2})^2\bigr)$.

Part (b). Setting $\exp(-2N(p - 1/2)^2) \le \delta$ and solving for N gives (3.5). □

Corollary 3.11 (Complexity–compute tradeoff)**.** *Suppose the base verification accuracy satisfies $p(F) > 1/2$ and depends on the composite complexity $C(F)$ through a decreasing function $p(F) = g(C(F))$ with $g(C) \to 1/2$ as $C \to \infty$. Then the number of self-consistency chains needed to reach accuracy $1 - \delta$ grows as $\Omega(1/(g(C(F)) - 1/2)^2)$, which diverges as $C(F) \to \infty$.*

Proof. By Theorem 3.10(b), $N \ge \ln(1/\delta)/(2(p(F) - 1/2)^2)$. Since $p(F) = g(C(F)) \to 1/2$ as $C(F) \to \infty$, the denominator vanishes, so $N \to \infty$. □

Remark 3.12. The Hoeffding bound is known to be loose for Bernoulli random variables; the exact majority probability can be computed via the binomial CDF. However, the Hoeffding bound correctly captures the *scaling*: $N = \Theta(1/(p - 1/2)^2)$ is tight up to constant factors. Our computational experiments confirm that the ordering of chain requirements across complexity levels is perfectly preserved ($\rho = 1.0$), even though absolute values differ.

Example 3.13. Consider two formulas: a Tseitin formula on a path graph with base accuracy $p = 0.90$ and a pigeonhole formula with $p = 0.60$. To reach 95%

majority vote accuracy ($\delta = 0.05$), Theorem 3.10 requires:

$$N_{\text{Tseitin}} \geq \frac{\ln 20}{2(0.40)^2} \approx 9.4 \implies N = 10,$$
$$N_{\text{PHP}} \geq \frac{\ln 20}{2(0.10)^2} \approx 149.8 \implies N = 150.$$

The pigeonhole formula requires roughly 15× more chains, reflecting its greater proof complexity.

3.4. **Phase transition in verification difficulty.** We provide a formal criterion for the existence of a phase transition in verification accuracy as a function of proof complexity.

Definition 3.14 (Piecewise linear model)**.** A *piecewise linear model* for verification accuracy as a function of composite complexity C is

$$p_{\max}(C) = \begin{cases} a_1 + b_1 \cdot C & \text{if } C \leq C^*, \\ a_2 + b_2 \cdot C & \text{if } C > C^*, \end{cases} \tag{3.6}$$

where C^* is the *critical threshold*, $b_1, b_2 < 0$ are degradation rates, and continuity at C^* imposes $a_1 + b_1 C^* = a_2 + b_2 C^*$.

Proposition 3.15 (Phase transition criterion)**.** *If the piecewise linear model* (3.6) *provides a statistically significant improvement over a single linear model* $p_{\max}(C) = a + bC$ *(as measured by an F-test with $p < 0.05$), and $|b_2| > |b_1|$, then we say a* phase transition in verification difficulty *occurs at C^*.*

Remark 3.16. This notion of phase transition is analogous to the well-known phase transition in random k-SAT, where the probability of satisfiability changes sharply at a critical clause-to-variable ratio [6]. In our setting, the "order parameter" is verification accuracy rather than satisfiability, and the control parameter is proof complexity rather than clause density.

3.5. **Computational verification.** We validate the theoretical predictions of Sections 3.1–3.4 on a dataset of 129 synthetic proof instances.

3.5.1. *Experimental setup.* We generate proof instances from four formula families with known complexity parameters:

TABLE 1. Summary of the 129 proof instances used for computational validation. The complexity range gives the minimum and maximum of the composite measure $C(F) = \log S(F) + \log w(F \vdash \bot) + \log d(F)$ within each family.

Family	**Count**	**Graph types / parameters**	$C(F)$ **range**
Tseitin	27	Path, cycle, grid, expander	$[1.6, 14.2]$
Pigeonhole	9	PHP_n^{n+1}, $n = 3, \ldots, 11$	$[5.3, 19.8]$
Pebbling	18	Chain, tree, pyramid DAGs	$[3.2, 18.5]$
Random 3-CNF	75	Ratios $\alpha \in [2.0, 5.0]$	$[1.0, 12.7]$

For each instance F, we compute exact complexity measures for the structured families and estimates for random 3-CNF. We then simulate the CoT verification

process (3.1) with 200–500 trials per instance, using a convergence rate model

$$\lambda(F) = \frac{\lambda_0}{1 + \sum_i w_i \cdot C_i(F)}, \tag{3.7}$$

where $C_i(F)$ are individual log-normalized complexity features and w_i are weights calibrated to published empirical data: $w_{\text{width}} = 0.35$, $w_{\text{space}} = 0.25$, $w_{\text{length}} = 0.25$, $w_{\text{depth}} = 0.15$, and $\lambda_0 = 2.0$. The calibration targets are: (i) single-pass accuracy of 94.5% and 64-vote accuracy of 97.5% from Heimdall [8]; (ii) difficulty-dependent saturation curves from Snell et al. [9].

3.5.2. *Results.*
Complexity–convergence correlation. Table 2 shows Spearman rank correlations between complexity features and convergence metrics.

Table 2. Spearman rank correlations between complexity features and convergence rate λ. All correlations are significant at the $p < 10^{-10}$ level after Bonferroni correction.

Feature	**Spearman** ρ	p**-value**
$C(F)$ (composite)	-0.991	$< 10^{-113}$
$\log S(F)$ (length)	-0.840	$< 10^{-35}$
$\log \mathrm{Sp}(F)$ (space)	-0.782	$< 10^{-27}$
$\log d(F)$ (depth)	-0.678	$< 10^{-18}$
$\log w(F \vdash \bot)$ (width)	-0.634	$< 10^{-15}$

The composite measure $C(F)$ achieves the strongest correlation ($\rho = -0.991$), consistent with Theorem 3.3(c). Among individual measures, proof length is the strongest single predictor, followed by space. Multiple linear regression on all complexity features yields $R^2 = 0.93$.
Accuracy–length relationship. The correlation between asymptotic accuracy $p_{\max}(F)$ and proof length is $\rho = -0.766$ ($p < 10^{-25}$), consistent with Theorem 3.6. Accuracy ranges from 0.972 for simple Tseitin formulas on path graphs to 0.880 for large pigeonhole instances.
Phase transition. Fitting the piecewise linear model (3.6) yields a critical threshold $C^* \approx 12.95$. The F-test comparing the piecewise model to a single linear fit gives $F = 7.24$ with $p = 0.008$. Below C^*, accuracy degrades at rate $b_1 = -0.0054$ per unit complexity; above C^*, the rate steepens to $b_2 = -0.0069$, a factor of 1.28. This confirms the phase transition criterion of Proposition 3.15.
Self-consistency chains. Table 3 compares empirically required chains with the Hoeffding prediction of Theorem 3.10.
Per-family comparison. Table 4 summarizes convergence metrics by formula family.

Pigeonhole formulas exhibit the lowest convergence rate and accuracy, consistent with their exponential proof complexity lower bounds [2]. Tseitin formulas on simple graphs are the easiest, reflecting their polynomial-size proofs on low-expansion graphs.

4. Discussion

4.1. **Connections to prior work.** Our framework unifies several threads from prior work.

TABLE 3. Number of independent chains needed to reach 95% majority vote accuracy. The Hoeffding bound is loose but preserves the ordering perfectly (Spearman $\rho = 1.0$).

Base accuracy p	**Chains (empirical)**	**Chains (Hoeffding)**
0.60	67	299
0.65	29	133
0.70	17	75
0.75	9	48
0.80	7	33
0.85	5	24
0.90	3	15

TABLE 4. Convergence metrics by formula family (mean $\pm$ std).

Family	**Conv. step**	**Conv. rate λ**	**Final accuracy**
Tseitin	4.9 ± 4.5	0.664 ± 0.259	0.940 ± 0.023
Pebbling	6.2 ± 4.8	0.612 ± 0.272	0.941 ± 0.021
Random 3-CNF	4.8 ± 4.5	0.570 ± 0.087	0.928 ± 0.016
Pigeonhole	6.2 ± 4.6	0.514 ± 0.176	0.923 ± 0.028

Test-time compute scaling. Snell et al. [9] show that compute-optimal scaling strategies are difficulty-dependent: easy problems saturate quickly, while hard problems require substantially more compute. Our results formalize this observation by identifying *what makes a problem hard* for verification: proof complexity measures, particularly the composite measure $C(F)$. Theorem 3.3 provides a formula relating the convergence step to $C(F)$, replacing the model-dependent pass@1 metric with a formula-intrinsic quantity.

Verification versus solving difficulty. Shi and Jin [8] demonstrate that verification difficulty does not correlate with problem-solving difficulty. Our framework explains this distinction: solving difficulty tracks search complexity (how hard it is to find a proof), while verification difficulty tracks proof complexity (how hard it is to check a given proof). These are related but distinct quantities—a problem may have a short proof that is hard to find, or a long proof that is easy to locate but hard to verify.

Width–size tradeoffs. Ben-Sasson and Wigderson's [1] width–size relation $w(F \vdash \bot) \leq w(F) + O(\sqrt{n \ln S(F)})$ parallels our finding that width and length jointly predict convergence better than either alone. The composite measure $C(F)$, which combines both, achieves $\rho = -0.991$, while individual measures achieve at most $\rho = -0.840$.

Space and practical hardness. Jarvisaälo et al. [4] show that space complexity predicts SAT solver hardness better than length or width. In our experiments, space ranks second ($\rho = -0.782$) after length ($\rho = -0.840$) in predicting convergence rate. This suggests that LLM verification may weight these measures differently than CDCL-based solvers, a distinction that merits empirical investigation.

Information-theoretic analysis. Ton et al. [10] prove that once an LLM encounters an unidentifiable sub-task in a CoT chain, all subsequent steps yield zero information gain. This "information plateau" corresponds to the convergence ceiling $p_{\max}(F)$ in our model: once verification accuracy saturates, no additional CoT steps improve performance. Moreover, their result strengthens our error accumulation bound (Theorem 3.6), since correlated errors make the bound optimistic.
Typed chain-of-thought. Perrier [7] maps CoT reasoning to typed proof terms via the Curry–Howard correspondence, providing a formal mechanism for measuring proof structure in natural language reasoning. The Minimal Path Size (MPS) metric introduced there is related to our proof depth measure $d(F)$. Combining Perrier's typed CoT framework with our complexity–convergence analysis could yield a unified theory of CoT verification for natural mathematical proofs.

4.2. **Implications.** Our results suggest several practical implications.
Compute allocation. Proof complexity measures can guide the allocation of test-time compute. Given a proof instance F, one can compute (or estimate) $C(F)$ and use Theorem 3.3 to predict the number of CoT steps and Corollary 3.11 to predict the number of self-consistency chains needed. This replaces the trial-and-error approach to test-time compute allocation with a principled, a priori estimate.
Selective human review. Instances with $C(F) > C^*$ fall in the harder regime identified by the phase transition (Proposition 3.15). These instances may benefit from human review or from structural decomposition into sub-proofs with lower individual complexity, rather than brute-force application of more CoT chains.
Fundamental limits. The exponential decay in accuracy with proof length (Theorem 3.6) implies that step-by-step verification of proofs with super-polynomial length faces a fundamental barrier. For formula families like pigeonhole, where any resolution proof has exponential length, CoT verification accuracy degrades to chance level for large instances (Example 3.9). Overcoming this barrier likely requires verification strategies that do not process each step independently, such as hierarchical verification or proof compression.

4.3. **Limitations.** We note several limitations of this work.

First, our theoretical framework is validated through simulation rather than direct LLM experiments. While the simulation is calibrated to published empirical data, the strong correlations obtained (particularly $\rho = -0.991$ for the composite measure) partly reflect the parametric model construction. The qualitative predictions—the ordering of difficulty across families and the existence of saturation behavior—are more robust than the specific correlation magnitudes.

Second, the step-independence assumption in Theorem 3.6 is a simplification. Real CoT traces exhibit sequential dependencies, and the information-theoretic analysis of Ton et al. [10] suggests that errors propagate in a correlated fashion.

Third, our complexity measures are defined for propositional resolution proofs. Extending these to first-order logic or to natural language mathematical proofs requires additional formalization, such as the typed CoT framework of Perrier [7].

Fourth, the convergence rate model (3.7) uses fixed weights w_i calibrated to aggregate statistics. These weights may vary across model architectures and scales.

4.4. **Open questions.** We conclude this section with several open questions.

Conjecture 4.1. *For at least one family among Tseitin, pigeonhole, and pebbling formulas, the Spearman rank correlation between $C(F)$ and the empirical convergence rate of GPT-4 or Claude verification exceeds -0.8 in absolute value.*

Conjecture 4.2. *For CoT verification by LLMs with bounded context windows, proof space $\mathrm{Sp}(F)$ is a stronger predictor of verification difficulty than proof length $S(F)$.*

Conjecture 4.3. *In the limit of large formula size n for random k-CNF at the satisfiability threshold, the phase transition in verification accuracy at C^* becomes sharp (i.e., the accuracy function is discontinuous in the limit).*

5. Conclusion

We have established a theoretical framework connecting classical proof complexity measures to the convergence behavior of chain-of-thought verification in large language models. Our main results show that: (1) the number of CoT steps to convergence scales as $O(\log(1/\varepsilon)/\lambda)$, where the convergence rate λ depends inversely on a composite proof complexity measure (Theorem 3.3); (2) asymptotic verification accuracy decays exponentially in proof length under step-independence (Theorem 3.6); and (3) self-consistency amplification requires $\Theta(1/(p-1/2)^2)$ chains, where the base accuracy p itself depends on proof complexity (Theorem 3.10). Computational experiments on 129 synthetic instances across four formula families strongly support these predictions, and a phase transition in verification difficulty is identified at a critical complexity threshold $C^* \approx 12.95$ (Proposition 3.15).

The central takeaway is that proof complexity theory provides the right lens for understanding the limits of neural proof verification. The framework offers a principled basis for allocating test-time compute in mathematical reasoning systems, replacing model-dependent difficulty estimates with formula-intrinsic complexity measures.

Two immediate directions for future work are: (i) empirical validation of the predicted complexity–convergence relationships using state-of-the-art LLMs, which would test Conjectures 4.1 and 4.2; and (ii) extension of the framework beyond propositional resolution to first-order and natural-language proofs, leveraging the Curry–Howard bridge introduced by Perrier [7].

References

1. Eli Ben-Sasson and Avi Wigderson, *Short proofs are narrow—resolution made simple*, Journal of the ACM **48** (2001), no. 2, 149–169.
2. Armin Haken, *The intractability of resolution*, Theoretical Computer Science **39** (1985), 297–308.
3. Wassily Hoeffding, *Probability inequalities for sums of bounded random variables*, Journal of the American Statistical Association **58** (1963), no. 301, 13–30.
4. Matti Jarvisaälo, Arie Matsliah, Jakob Nordström, and Stanislav Živný, *Relating proof complexity measures and practical hardness of SAT*, Proceedings of the 18th International Conference on Principles and Practice of Constraint Programming (CP), Springer, 2012, pp. 316–331.
5. Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and OpenAI Alignment, *Let's verify step by step*, International Conference on Learning Representations (ICLR), 2024.
6. Marc Mézard, Giorgio Parisi, and Riccardo Zecchina, *Analytic and algorithmic solution of random satisfiability problems*, Science **297** (2002), no. 5582, 812–815.
7. Guillaume Perrier, *Typed chain-of-thought: a Curry–Howard framework for verifying LLM reasoning*, arXiv preprint arXiv:2510.01069 (2025).

8. Zhangchen Shi and Shuai Jin, *Heimdall: test-time scaling on generative verification*, arXiv preprint arXiv:2504.10337 (2025).
9. Charlie Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar, *Scaling LLM test-time compute optimally can be more effective than scaling model parameters*, arXiv preprint arXiv:2408.03314 (2024).
10. Jean-Francois Ton, Muhammad Faaiz Taufiq, and Yang Liu, *Understanding chain-of-thought in LLMs through information theory*, International Conference on Machine Learning (ICML), 2025.
11. Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou, *Self-consistency improves chain of thought reasoning in language models*, International Conference on Learning Representations (ICLR), 2023.
12. Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou, *Chain-of-thought prompting elicits reasoning in large language models*, Advances in Neural Information Processing Systems (NeurIPS), vol. 35, 2022, pp. 24824–24837.